



Tribhuvan University

Faculty of Humanities and Social Sciences

Sabaiko Blood Bank: A Blood Donation Platform

A PROJECT REPORT

Submitted to

Department of Computer Application

Nepathya College

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Under the supervision of

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Supervisor's recommendation

I hereby recommend that this project prepared under my supervision by Dinesh Poudel entitled “**Blood Donation Platform: Sabaiko Blood Bank**” in partial fulfilment of the requirements for the degree of Bachelor of Computer Application is recommended for proposal evaluation.

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LETTER OF APPROVAL

This document certifies that Dinesh Poudel's project, "**Sabaiko Blood Bank**" has undergone a comprehensive evaluation to fulfil the requirements for the degree of Bachelor in Computer Application. In our expert opinion, this project demonstrates satisfactory scope and exceptional quality, meeting the standards expected for the degree.

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Abstract

Blood transfusion is a vital medical service required in emergencies, surgeries, and the treatment of chronic diseases. In Nepal, the lack of a centralized and real-time donor registry often causes delays in locating compatible blood donors. Hospitals and families frequently rely on informal communication channels, which are inefficient and unreliable during critical situations.

“Sabaiko Blood Bank” is a web-based blood donation platform developed to provide a secure and centralized system that connects verified donors with receivers such as hospitals and organizations. The platform allows donors to register, update availability, and maintain donation history, while receivers can search donors based on blood group and detailed Nepal-specific location filters. Administrative verification, role-based access control, and encrypted authentication ensure system security and reliability.

Developed using Spring Boot, Spring Security, Thymeleaf, and PostgreSQL, the system offers a responsive and scalable solution. The platform aims to reduce donor search time and improve transparency, efficiency, and accessibility in Nepal’s blood donation process.

Keywords: Blood Donation Platform, Sabaiko Blood Bank, Donor Management, Healthcare System, Nepal.

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List of Abbreviations

1. API: Application Programming Interface
2. CASE: Computer-Aided Software Engineering
3. CRUD: Create, Read, Update, Delete
4. CSS: Cascading Style Sheets
5. DFD: Data Flow Diagram
6. ER: Entity Relationship
7. HTML: Hyper Text Markup Language
8. IDE: Integrated Development Environment
9. SQL: Structured Query Language
10. UI: User Interface

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Chapter 1: Introduction

1.1 Introduction

Blood transfusion is a critical medical procedure essential for saving lives in emergencies, surgeries, trauma cases, and chronic illnesses such as thalassemia and cancer. In Nepal, despite efforts by the Nepal Red Cross Society and various hospitals, the demand for blood frequently exceeds supply, particularly during festivals, natural disasters, or in remote areas. The absence of a centralized, real-time donor registry forces families and healthcare providers to rely on fragmented social media groups, WhatsApp broadcasts, and phone calls, often leading to unverified information, significant delays, and preventable loss of life.

Sabaiko Blood Bank is a web-based application developed to overcome these challenges by providing a secure, centralized platform that connects verified donors with receivers (hospitals, blood banks, and organizations). The system enables donors to register and update availability, allows receivers to search compatible donors by blood group and precise location (province to ward level), The system ensures accessibility and brings much-needed transparency, speed, and trust to Nepal's blood donation process. By employing Spring Boot, a robust Java based framework, and Thyme leaf, a powerful templating engine for creating dynamic web pages, the system ensures a seamless user experience.

1.2 Problem Statement:

The blood donation ecosystem in Nepal remains highly fragmented and inefficient, posing significant risks during medical emergencies. Hospitals and blood banks typically maintain isolated donor databases with no standardized mechanism for inter-hospital data sharing or nationwide access. This silos information, making it extremely difficult for healthcare providers to quickly identify available donors when urgent needs arise, such as in trauma cases, surgeries, or natural disasters. Furthermore, the absence of a centralized registry means there is no reliable way to track donors' real-time availability status, blood group compatibility, or last donation date, forcing families and medical staff to rely on outdated lists or personal networks.

Compounding these issues, urgent blood requirements are frequently broadcast through informal channels like Facebook groups, WhatsApp messages, and social media posts. While these platforms enable rapid dissemination, they often spread unverified or inaccurate information, leading to misinformation, fake donor claims, and wasted time chasing unreliable leads. Manual verification of donor health eligibility—such as confirming recent donation history or medical fitness—adds further delays, consuming critical hours when minutes matter. In remote or rural areas, geographic barriers exacerbate the problem, as locating nearby donors becomes nearly impossible without structured location data.

These systemic gaps result in preventable delays and, tragically, loss of lives that could have been saved with timely transfusions. **Sabaiko Blood Bank** directly addresses these challenges by introducing a secure, centralized, real-time platform that verifies donors through admin oversight, enables precise location-based searching (down to ward level), tracks availability, and facilitates efficient matching between donors and receivers (hospitals or organizations), ultimately bringing transparency, speed, and reliability to Nepal's blood donation process.

1.3 Objectives

- To develop a centralized and secure registration system for blood donors and receivers with mandatory admin verification to ensure authenticity and trust.
- To develop secured centralized registration system for blood donors.
- To create a searchable and filterable donor database that allows receivers to find donors by blood group, province, district, municipality (palika), ward, and real-time availability.
- To implement a structured donation tracking system that records donation history and supports a two-step verification process by receivers and administrators.
- To provide a robust administrative panel for approving users, managing roles and statuses, monitoring donations, and maintaining overall system integrity.
- To ensure strong security through role-based access control, encrypted authentication, and status-based login restrictions.
- To design a responsive, mobile-friendly, and user-friendly interface accessible across all regions of Nepal on various devices.

1.4 Scope and Limitation

Scope

- Covers donor registration, recipient request creation, automated matching, notification, and admin management.
- Supports all eight blood types (A+, A-, B+, B-, AB+, AB-, O+, O-).
- Location-based search at Province to Ward level.
- Deployable on free-tier cloud services for initial nationwide use.

Limitation

- Does not integrate directly with blood-bank inventory systems of hospitals (future scope).
- SMS notification relies on third-party gateways (currently implemented via phone number only in prototype).
- Offline coordination between donor and recipient after matching is manual.
- Does not perform medical screening; donor self-declaration is used.

1.5 Development Methodology

Waterfall Methodology is a linear and sequential software development lifecycle (SDLC) model where each phase of the process flows downwards, like a waterfall, into the next phase. It is one of the oldest and most traditional models used in software engineering. This methodology is suitable for project as it provides clear and well-defined structure and through documentations.

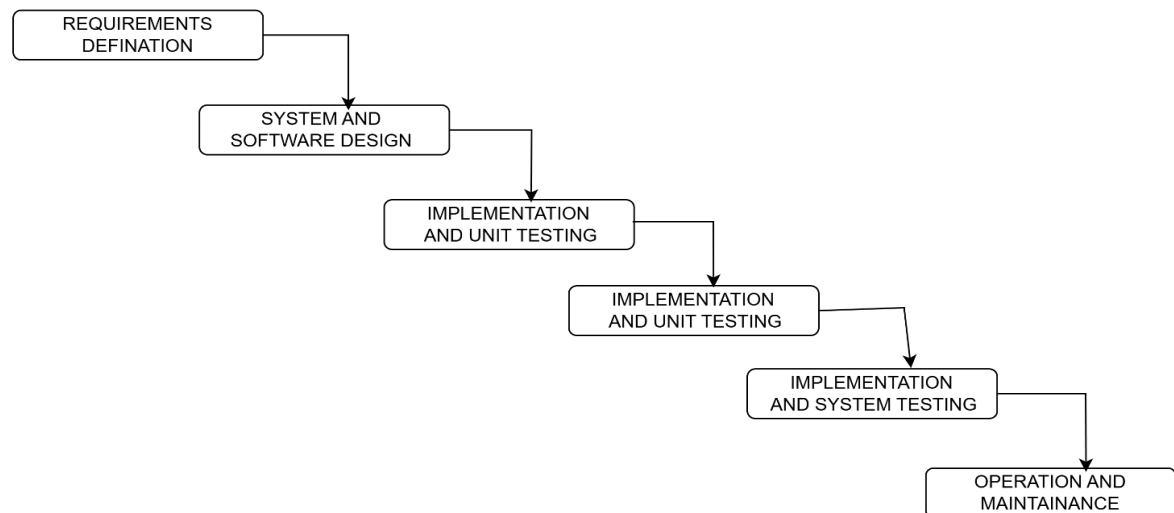


Figure 1: Waterfall method

1.6 Report Organization

The report is organized into several key sections that together provide a clear and comprehensive understanding of the project. It begins with an introduction that outlines the project's objectives, problem statement, scope, and limitations, establishing the foundation for the work. This is followed by a background study that explains the essential theories, terminologies, and related literature, helping to contextualize the project within existing research and similar systems. The next section focuses on system analysis and design, detailing both functional and non-functional requirements, along with system diagrams such as use case representations, feasibility studies, and the overall system design. Finally, the report concludes with a summary of the project's outcomes, the challenges encountered during development, and practical recommendations for future improvements.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Blood donation is a vital healthcare service that supports emergency treatment, surgeries, and management of chronic diseases. In Nepal, blood donation activities are mainly handled by the Nepal Red Cross Society and hospital-based blood banks. However, the existing system faces challenges such as the absence of a centralized donor database, difficulty in finding available donors quickly, geographic limitations, and lack of proper verification mechanisms. These issues often lead to delays, especially during emergencies and high-demand periods.

Traditional blood donation coordination relies heavily on manual processes, phone calls, and informal communication channels, which are inefficient and unreliable. There is no unified platform to systematically connect verified donors with hospitals or organizations in need, resulting in time-consuming searches and increased risk for patients.

Sabaiko Blood Bank is a web-based blood donation management platform developed to address these challenges. The system provides a centralized and secure platform where verified donors and receivers can connect efficiently. It enables donor search based on blood group and Nepal-specific address hierarchy, ensuring accurate location-based matching.

By implementing role-based access control, admin verification, and donation history tracking, the platform ensures trust, security, and transparency. Through the use of modern web technologies, Sabaiko Blood Bank aims to improve the efficiency, reliability, and accessibility of blood donation services in Nepal.

2.2 Literature Review

Several studies and institutional efforts have focused on improving blood donation and blood bank management through information systems. However, existing solutions often emphasize inventory management over donor–receiver connectivity and lack strong verification and location-based search mechanisms.

Sharma et al. [1] proposed a Blood Bank Management System using PHP and MySQL, primarily designed for hospital-level blood inventory management. While the system improves internal record keeping and stock monitoring, it does not support public donor search, real-time donor availability, or location-based filtering, which limits its effectiveness during emergency blood requirements.

The Ministry of Health and Population, Nepal [2], through the National Blood Transfusion Service Guidelines, emphasizes the importance of donor safety, traceability, and proper verification in blood transfusion services. However, these guidelines focus more on operational and regulatory aspects rather than providing a digital platform for donor discovery, tracking, and coordination.

The Nepal Red Cross Society Annual Report 2023–2024 [5] highlights ongoing challenges in meeting blood demand, particularly during emergencies and high-demand periods. Although the organization plays a central role in blood collection and distribution, the report indicates a need for improved digital systems to enhance coordination, accessibility, and timely donor identification.

From a technological perspective, modern web frameworks such as Spring Boot with Spring Security [3] and React [4] provide robust support for building secure, scalable, and user-friendly web applications.

Based on the limitations identified in existing systems and institutional reports, the present work, Sabaiko Blood Bank, aims to bridge the gap by providing a centralized, secure, and location-aware blood donation management platform. The system integrates strong admin verification, detailed Nepal-specific address filtering, and a modern web technology stack to better support real-world blood donation needs in Nepal.

Chapter 3: System Analysis and Design

3.1 System Analysis

System Analysis refers into the process of examining a system with the intent of improving it through better procedures and methods. It is the process of planning a new system to either replace or complement an existing system. It is therefore, the process of gathering and interpreting facts, diagnosing problems and using the information to recommend improvements in the system.

3.1.1 Requirement Analysis

Requirement analysis involves refining and prioritizing the collected requirements to ensure they align with the project's objectives and constraints. This process includes:

i. Functional Requirements

Functional requirements define what a system should do. They describe the specific behavior, features, and functionalities that the system must provide. These requirements represent the actions the system performs in response to user inputs or system conditions. The functional requirements of the Blood Donation Platform include:

- **User Authentication:** The system shall allow users (donors, receivers, and admins) to register and log in using a username and password. The system shall validate login credentials and deny access if the credentials are incorrect or the user is not approved.
- **Donor Registration and Profile Management:** The system shall allow donors to register and manage their profile information, including blood group, address, medical details, and availability status.
- **Receiver Registration:** The system shall allow hospitals or receivers to register by providing organizational details and verification information.
- **Donor Search Functionality:** The system shall allow receivers to search for donors based on blood group, district, municipality, ward number, and availability status.

- **Blood Donation Request Handling:** The system shall allow receivers to search and contact suitable donors for urgent blood requirements.
- **Admin Verification and User Management:** The system shall allow the admin to verify donor and receiver documents, approve or reject user registrations, restrict users, and manage system users.
- **Dashboard and Reporting:** The system shall provide an admin dashboard displaying system statistics such as total donors, receivers, and pending approvals.
- **Password Recovery:** The system shall allow users to recover forgotten passwords through a secure email-based password reset process.



Figure 2: Use case diagram

ii. Non-Functional Requirements

Non-functional requirements define how the system performs a particular function, focusing on overall quality attributes such as performance, scalability, security, usability, and reliability. These requirements describe the operational characteristics of the system rather than specific behaviors. The non-functional requirements of the Blood Donation Platform include:

- **Performance:** The system shall respond to donor search queries and dashboard requests within 2 seconds under normal operating conditions.
- **Scalability:** The system shall support at least 500 concurrent users and allow future expansion in terms of users, data volume, and additional features without requiring major architectural changes.
- **Security:** User passwords shall be securely hashed using the BCrypt algorithm. Role-based access control shall be enforced to prevent unauthorized access to sensitive system resources.
- **Availability:** The system shall maintain at least 99% uptime when deployed on free-tier or entry-level hosting platforms, excluding scheduled maintenance periods.
- **Usability:** The system shall provide a responsive and user-friendly interface that is fully accessible on mobile and desktop web browsers, enabling users to perform basic operations with minimal learning effort.
- **Maintainability:** The system shall be designed using a modular architecture with proper documentation, allowing updates, bug fixes, and enhancements to be applied with minimal downtime.
- **Compatibility:** The system shall be compatible with major modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

3.1.2 Feasibility Analysis

Technical Feasibility: The system is technically feasible using Spring Boot, PostgreSQL, and Thymeleaf, which are stable, well-documented, and widely used technologies. Free-tier platforms like Railway and Vercel support smooth deployment.

Operational Feasibility: The system is easy to use and requires minimal training. Its intuitive interface and support for Nepali-language labels make it suitable for quick adoption.

Economic Feasibility: The project is cost-effective as it uses open-source tools and free-tier hosting, resulting in zero development and deployment cost.

Schedule Feasibility: The Event Management System (EMS) project is schedule-feasible, as the development process is broken down into clearly defined phases. The project timeline is realistic, with the initial phases (requirement analysis and system design) estimated to take 3-4 weeks. The development phase, including both frontend and backend, is expected to take 4-6 weeks, followed by 1-2 weeks for testing and debugging. Deployment and go-live can be completed in 1-2 weeks. Given the familiarity with the chosen technologies (Spring Boot, PostgreSQL, and Thymeleaf), the project is expected to be completed within the planned schedule without significant delays.

Project Timeline

Gantt Chart Visualization

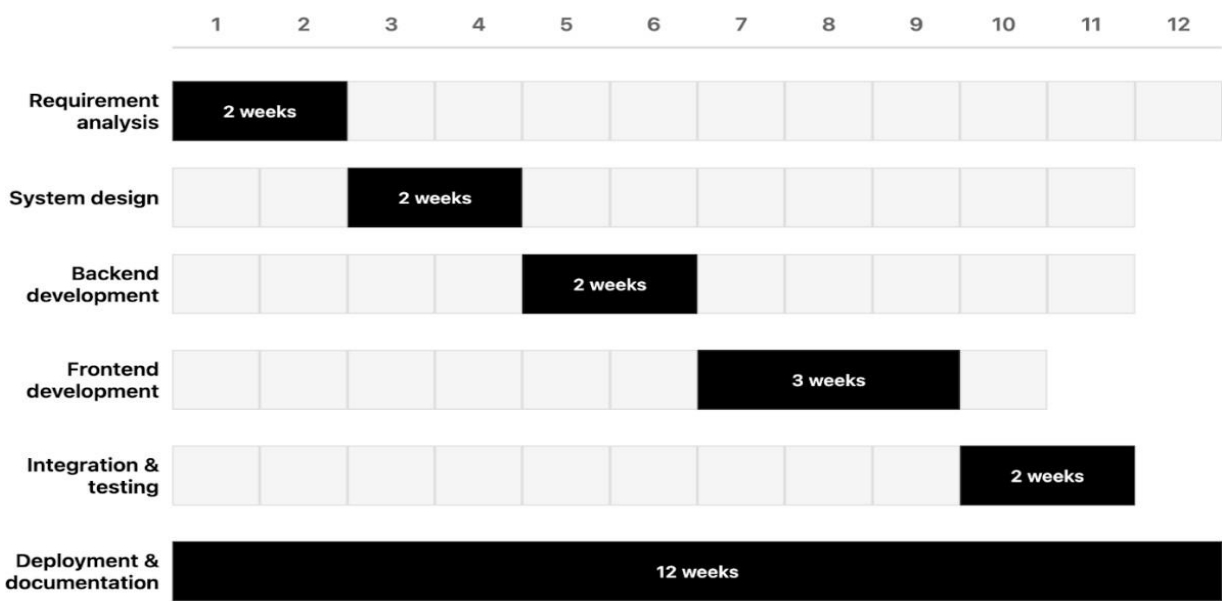


Figure 3: Gantt Chart for Blood Donation Platform

3.1.3 Data Modelling (ER-Diagram)

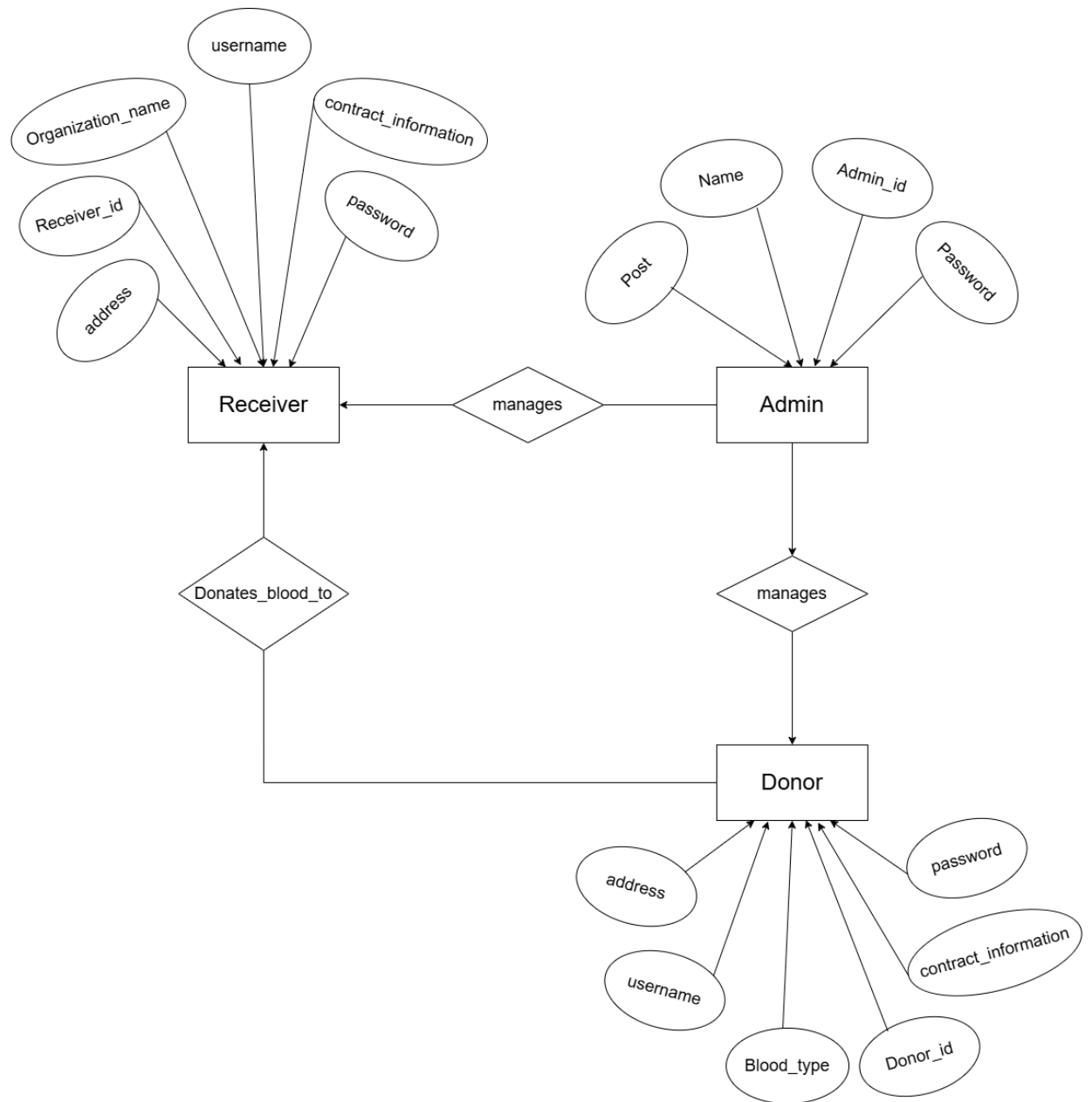


Figure 4: ER diagram

3.1.4 Process Modelling (DFD)

Process Modelling with DFDs visually represents data flow in a system, showing processes, data stores, and external entities. Level 0 gives a high-level overview, Level 1 breaks it into major processes, and Level 2 details each process. They help understand, design, and document systems effectively.

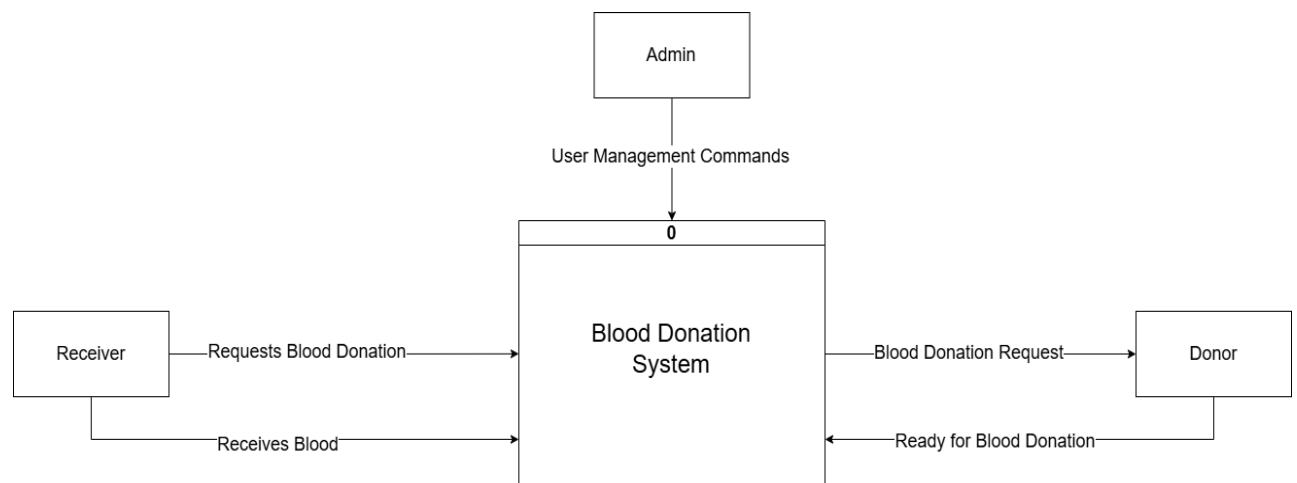


Figure 5: Context Level DFD

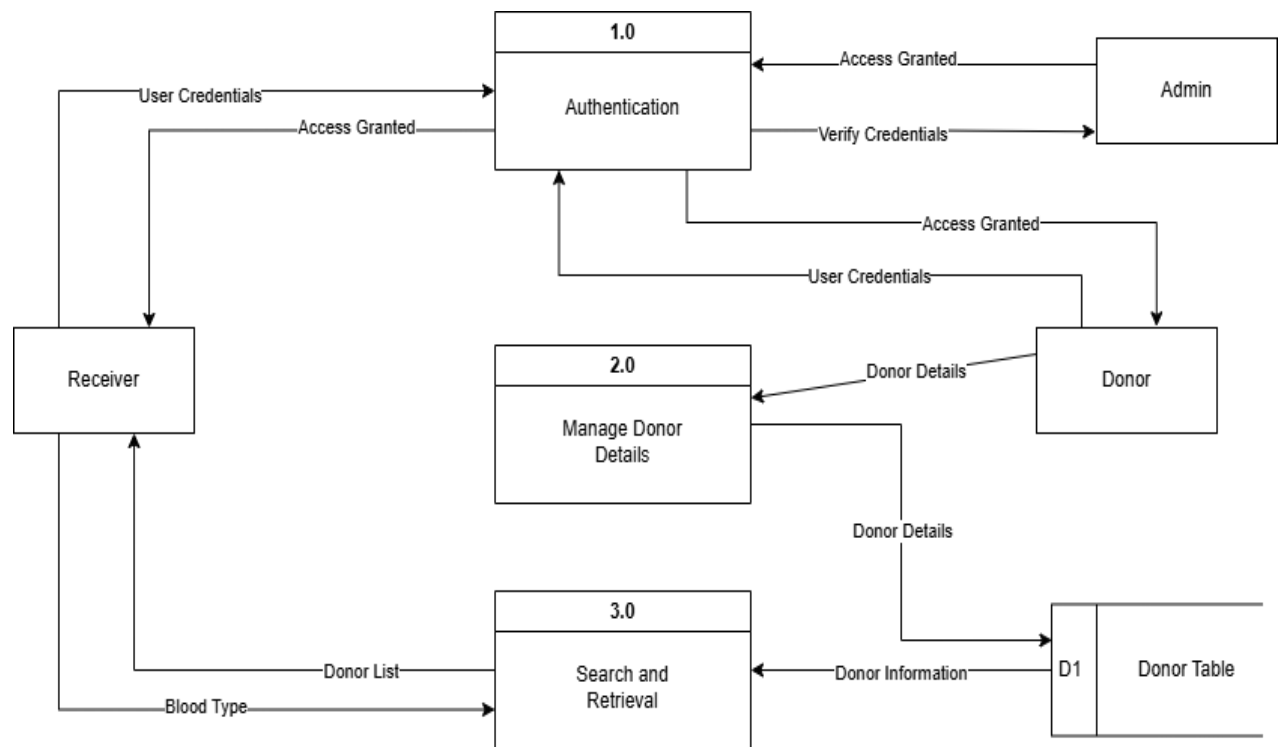


Figure 6: Level-1 DFD

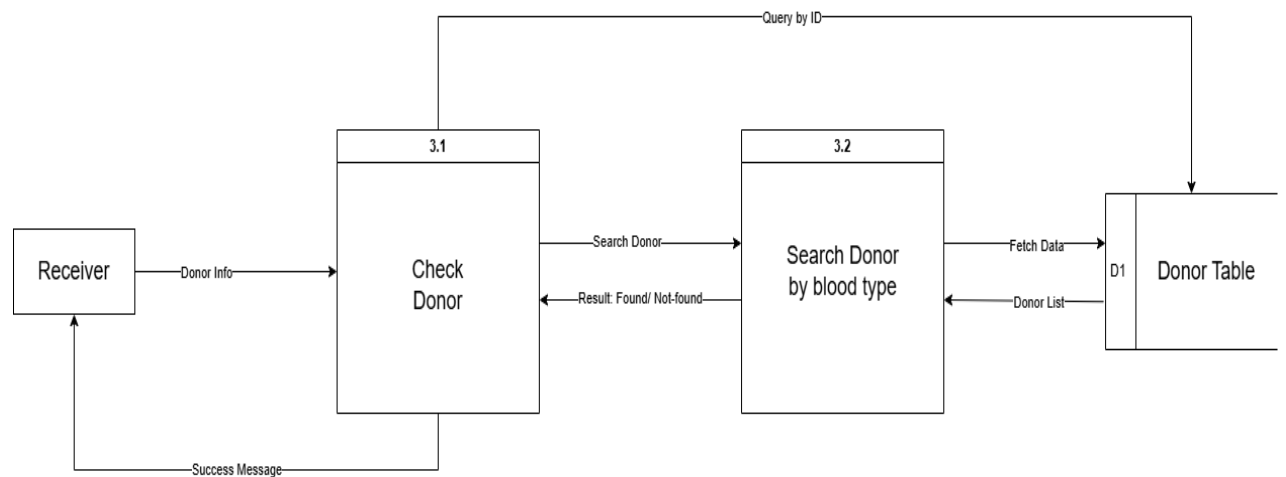


Figure 7: Level-2 DFD

3.2 System Design

3.2.1 Architectural Design

Architectural design is a process for identifying the sub-systems making up a system and the framework for sub-system control and communication.

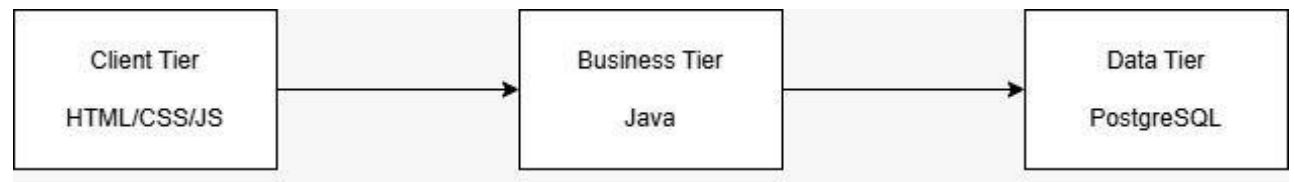


Figure 8: Architectural Design

3.2.2 Database Schema Design

The database schema for **Sabaiko Blood Bank** has been carefully designed following relational database principles to ensure data integrity, security, scalability, and maintainability. PostgreSQL is used as the database management system due to its ACID compliance, robustness, and strong support for constraints and indexing mechanisms.

The schema is normalized to eliminate redundancy and maintain consistency across related entities. Core entities in the system include Users, Donor Details, Receiver Details, Donation History, and Password Reset Tokens. Proper foreign key constraints, unique constraints, and default values are implemented to enforce referential integrity and business rules.

1. Users Table

The Users table serves as the central authentication and authorization entity of the system. It stores login credentials and role-based access information for all system users, including Admins, Donors, and Receivers.

Each user is assigned a specific role and status. The role determines access privileges within the system, while the status controls whether the user is allowed to log in (e.g., PENDING, VERIFIED, RESTRICTED).

Table 1: Users table

Column Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Unique identifier for each user
username	VARCHAR(255)	UNIQUE, NOT NULL	Unique login username
email	VARCHAR(255)	UNIQUE, NOT NULL	User's registered email address
password	VARCHAR(255)	NOT NULL	BCrypt-encrypted password
role	VARCHAR(20)	NOT NULL	User role (ADMIN, DONOR, RECEIVER)
status	VARCHAR(20)	NOT NULL	Account status (PENDING, VERIFIED, RESTRICTED)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
updated_at	TIMESTAMP	—	Last update timestamp

Security is enforced through password hashing using BCrypt, and role-based access control is implemented via Spring Security at the application level.

2. Donor Details Table

The Donor Details table stores donor-specific information separate from authentication data. This separation ensures proper normalization and enhances security by isolating login credentials from personal and medical data.

Each donor record maintains a one-to-one relationship with the Users table through the user_id foreign key.

Table 2: Donor Details Table

Column Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Unique donor record identifier
user_id	INTEGER	FK → users(id), UNIQUE	Associated user account
full_name	VARCHAR(255)	NOT NULL	Donor's full name
blood_group	VARCHAR(5)	NOT NULL	Blood group (A+, A-, B+, B-, AB+, AB-, O+, O-)
phone_number	VARCHAR(20)	NOT NULL	Contact number
province	VARCHAR(100)	NOT NULL	Province name
district	VARCHAR(100)	NOT NULL	District name
municipality	VARCHAR(150)	NOT NULL	Municipality/Palika
ward_number	INTEGER	NOT NULL	Ward number
last_donation_date	DATE	—	Most recent donation date
available	BOOLEAN	DEFAULT TRUE	Availability status for donation

medical_history	TEXT	—	Optional medical notes
profile_image	VARCHAR(255)	—	File path to profile image
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	—	Record update timestamp

The blood_group field is immutable after registration to maintain medical consistency. The availability flag allows real-time filtering during donor search.

3. Receiver Details Table

The Receiver Details table stores information about hospitals, blood banks, and organizations that request blood donations. Similar to donors, receiver information is stored separately from authentication credentials to maintain clean separation of concerns.

Each receiver record is linked one-to-one with the Users table.

Table 3: Receiver details table

Column Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Unique receiver identifier
user_id	INTEGER	FK → users(id), UNIQUE	Associated user account
organization_name	VARCHAR(255)	NOT NULL	Name of hospital or organization
contact_person	VARCHAR(255)	NOT NULL	Responsible authority
phone_number	VARCHAR(20)	NOT NULL	Official contact

			number
province	VARCHAR(100)	NOT NULL	Province name
district	VARCHAR(100)	NOT NULL	District name
municipality	VARCHAR(150)	NOT NULL	Municipality/Palika
ward_number	INTEGER	NOT NULL	Ward number
description	TEXT	—	Additional organizational details
profile_image	VARCHAR(255)	—	Logo or profile image path
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration timestamp
updated_at	TIMESTAMP	—	Last update timestamp

3.2.3 Interface Design

The system interface of *Sabaiko Blood Bank* is designed to be simple, user-friendly, and role-based. The interface ensures smooth navigation and secure access for Admin, Donor, and Receiver users. It is responsive and accessible across desktop and mobile devices.

The system provides the following main interfaces:

- **Login Page:**

Allows users (Admin, Donor, Receiver) to log in using their username and password. The system validates credentials and redirects users to their respective dashboards.

- **Registration Forms:**

Separate forms for Donor and Receiver registration. Users enter personal or organizational details, which require admin approval before activation.

- **Donor Dashboard:**

Allows donors to update profile information, manage availability status, and view donation history.

- **Receiver Dashboard:**

Enables receivers to search donors by blood group and location, view donor details, and verify donations.

- **Public Search Page:**

Allows non-registered users to search available donors by blood group and location, with limited information visibility.

- **Admin Dashboard:**

Allows the administrator to approve users, manage accounts, monitor donation records, and control system activities.

The interface is designed with clear navigation, consistent layout, and secure role-based access to ensure efficiency and ease of use.

Chapter 4: Implementation and Testing

4.1 Implementation

In this phase, the theoretical design is turned into a working system. The design phase has been completed and now it is time for the technical implementation. This phase includes coding the application according to the requirements and specifications of the Sabaiko Blood Bank system. The system connects verified blood donors with receivers across Nepal through a secure, role-based web platform built using the Spring Boot framework.

4.1.1 Tools Used

Front End:

The frontend is the part of the web application that users interact with directly. In this project, the following frontend tools are used:

- iii. **HTML5:** To define the structure of the web pages.
- iv. **CSS3 with Tailwind CSS and Bootstrap 5:** For styling HTML pages with a responsive, mobile-first design and consistent UI components.
- v. **Thymeleaf:** For server-side rendering, loading dynamic content to the frontend from the backend, and applying role-based view logic within HTML templates.
- vi. **JavaScript (ES6+):** For adding interactive features to the frontend, such as form validation and dynamic UI behaviour.

Back End:

The backend part of the application encompasses the business logic, data processing, and database interaction that occurs away from the user. It manages the data, processes requests from the frontend, and sends appropriate responses. The following backend tools and frameworks are used in this system:

- vii. **Spring Boot 3.3.5 (Spring Security, Spring Data JPA, Spring MVC):** Spring Boot was chosen as the backend framework due to its robust features for building enterprise-level applications. Its auto-configuration capability significantly reduces development time. Spring Security handles role-based access control and

- BCrypt password encryption, while Spring Data JPA manages all database interactions through the Hibernate ORM.
- viii. **Java 21:** The primary programming language used for all backend logic, chosen for its stability, strong typing, and extensive ecosystem.
 - ix. **PostgreSQL 12+:** Selected as the database system for its reliability and powerful features. Its ACID compliance ensures data integrity, which is critical for storing sensitive donor and receiver information. Managed using pgAdmin.
 - x. **Gradle:** Used as the build tool for dependency management and build automation.
 - xi. **Git and GitHub:** Used for version control and source code management throughout the development process.

4.1.2 Implementation Details of Modules

The application is divided into four main modules: Authentication and User Management, Donor Management, Receiver Management and Donor Search, and Admin Management. Each module is implemented using Spring Boot controllers, with services handling the business logic and repositories managing data access.

1. Authentication and User Management Module:

The Authentication and User Management module handles all user-related operations including registration, login, logout, and profile management. It uses Spring Security with BCrypt password encryption for secure credential storage. The module supports three distinct user roles (Admin, Donor, Receiver) and enforces a status-based login policy, where only users with a VERIFIED status can access the system. New registrations are assigned a PENDING status and must be approved by an admin before gaining access.

Example:

- **registerDonor() / registerReceiver():** Validates the submitted registration form, encrypts the password using BCrypt, creates a User record with PENDING status, and persists the corresponding DonorDetails or ReceiverDetails entity.
- **loadUserByUsername():** Custom UserDetailsService implementation that retrieves the user record from the database during login and checks that the user's status is VERIFIED before allowing access.

- **onAuthenticationSuccess():** Custom success handler that reads the authenticated user's role and redirects them to their respective dashboard (Admin, Donor, or Receiver) after successful login.
- **deleteAccount():** Allows donors and receivers to permanently delete their own accounts. Clears the security session and redirects the user to the login page with a confirmation parameter.

2. Donor Management Module:

The Donor Management module allows donors to manage their profiles, update their availability status, and upload a profile image. Donor-specific data such as blood group, Nepal-specific address details (province, district, palika, ward), medical history, and biography are stored in a dedicated DonorDetails entity linked one-to-one with the User entity. Notably, blood group is immutable after registration to maintain medical data integrity.

Example:

- **updateDonorProfile():** Accepts updated profile fields from the DonorUpdateDTO, validates the data, saves changes to the DonorDetails record, and redirects to the profile view page.
- **uploadProfileImage():** Receives the uploaded image file via MultipartFile, validates the size (max 10 MB) and type, stores the file in the uploads/profiles/ directory, and saves the file path to the donor's record in the database.
- **searchReceivers():** Allows donors to search for verified receiver organisations by phone number, returning matching results from the ReceiverDetails repository.

3. Receiver Management and Donor Search Module:

This module enables receivers (hospitals, blood banks, and organisations) to manage their profiles and search for available blood donors. The core feature is the proximity-sorted donor search, which prioritises results based on geographic closeness to the receiver's registered address. A public donor search is also accessible without login through the SearchController. Verified donations are tracked through the DonationHistory entity with a two-level verification process involving both the receiver and the admin.

Example:

- **searchDonors():** Accepts a blood group filter from the receiver, queries the DonorDetailsRepository for available donors, and sorts results by proximity: same province and district first, same province second, and all others last.
- **publicSearchDonors():** Provides the public /search endpoint accessible without login. Filters donors by blood group, province, district, palika, and ward number, and displays only available donors without exposing contact details.
- **verifyDonation():** Called by the receiver after a blood donation has taken place. Creates a DonationHistory record linking the donor and receiver, sets verifiedByReceiver to true, and records the donation date.
- **updateReceiverProfile():** Accepts updated profile fields from the ReceiverUpdateDTO, persists changes to the ReceiverDetails record, and handles profile image uploads in the same manner as the donor module.

4. Admin Management Module:

The Admin Management module provides a centralised dashboard for system oversight and user governance. This is the primary control layer that ensures only legitimate users access the platform. The admin can review pending registrations with full profile details, approve or reject new users, restrict or unrestrict existing accounts, and permanently delete users. A default admin account is created automatically by the DataInitializer component on first application run.

Example:

- **approveUser():** Updates the target user's status from PENDING to VERIFIED, granting them login access to the system.
- **rejectUser():** Permanently removes the pending user's record from the database, denying them access to the platform.
- **restrictUser():** Sets a verified user's status to RESTRICTED, blocking their login immediately without deleting their account or data.
- **deleteUser():** Permanently deletes a user and all associated data from the database, removing them from all listings and searches.

4.2 Testing

Testing is a critical phase in the development process to ensure the application functions as intended. It involves verifying both individual components (unit testing) and the system as a whole (system testing). The primary objective of test case design is to derive a set of tests that has the highest likelihood of uncovering defects in the software.

4.2.1 Test Cases for Unit Testing

Unit testing was performed on individual service and controller methods to ensure each component behaves correctly in isolation.

Table 4: Authentication and User Management Unit Test

Test Case ID	Test Case	Input	Expected Result	Status
A1	Verify that registerDonor() correctly registers a new donor	A valid RegistrationDTO object with donor details and blood group	User saved in database with PENDING status; DonorDetails record linked to User	Pass
A2	Verify that registration is rejected when a duplicate username is submitted	A username that already exists in the database	Exception thrown; registration rejected with a duplicate username error message	Pass
A3	Verify that loadUserByUsername() blocks login for a PENDING status user	A valid username and password of a PENDING user	Authentication exception thrown; user denied login and redirected to login page	Pass

Table 5: Donor Management Unit Test

Test Case ID	Test Case	Input	Expected Result	Status
D1	Verify that updateDonorProfile() correctly saves updated donor profile fields	A valid DonorUpdateDTO with updated address and availability fields	DonorDetails record updated in database; profile view page displayed with new values	Pass
D2	Verify that uploadProfileImage() rejects a file exceeding the 10 MB size limit	An image file larger than 10 MB	Upload rejected; error message returned; no file written to the uploads directory	Pass

Table 6: Receiver Management and Donor Search Unit Test

Test Case ID	Test Case	Input	Expected Result	Status
R1	Verify that searchDonors() returns results sorted by proximity to the receiver's address	Blood group filter B+ with donors in same district, same province, and other provinces	Results returned in order: same district first, same province second, all others last	Pass
R2	Verify that verifyDonation() creates a correct DonationHistory record	A valid donor ID submitted by an authenticated receiver	DonationHistory record created with verifiedByReceiver set to true and correct	Pass

Test Case ID	Test Case	Input	Expected Result	Status
			donation date	

Table 7: Admin Management Unit Test

Test Case ID	Test Case	Input	Expected Result	Status
AD1	Verify that approveUser() changes a user's status from PENDING to VERIFIED	A valid user ID of a PENDING user	User status updated to VERIFIED; user can now log in to the system	Pass
AD2	Verify that restrictUser() blocks a VERIFIED user from logging in	A valid user ID of a VERIFIED user	User status set to RESTRICTED; login denied immediately without account deletion	Pass

4.2.2 Test Cases for System Testing

System testing is a type of software testing that verifies the integrated system meets its requirements and performs as expected under realistic conditions. The following test cases cover the complete end-to-end workflows of the Sabaiko Blood Bank platform.

Table 8: System Testing

Test Case ID	Test Case	Input	Expected Result	Status
S1	Donor registers successfully	Donor provides valid details including blood group and address, then clicks “Register”	User saved in database with PENDING status; DonorDetails record created and linked	Pass
S2	Admin approves a pending user; user logs in with valid credentials	Admin clicks “Approve” on the pending user; user enters correct username and password and clicks “Login”	User status updated to VERIFIED; authenticated session established; user redirected to their role-specific dashboard	Pass
S3	Receiver searches for available blood donors by blood group	Receiver navigates to donor search, selects blood group A+, and submits the search form	Available donors displayed, sorted by proximity to the receiver’s address; donor contact details visible	Pass
S4	Receiver verifies a blood donation after it has taken place	Receiver submits donor ID via the verify-donation form after blood collection	DonationHistory record saved in database with verifiedByReceiver = true; donation visible in history for both donor and receiver	Pass
S5	Unauthenticated user searches for donors on the public search page	User visits /search, selects a blood group and province, and clicks Search	Matching available donors displayed without requiring login; sensitive contact details not exposed to unauthenticated users	Pass
S6	User tries to log in with invalid credentials	User enters an incorrect username or password and clicks “Login”	Error message displayed on the login page; session not established; user remains unauthenticated	Pass
S7	A logged-in donor attempts to access a receiver-only page	Donor navigates directly to /receiver/dashboard while authenticated	Access denied by Spring Security; user redirected to an error page due to insufficient role permissions	Pass
S8	Admin deletes a verified user account from the	Admin navigates to the verified users list and clicks “Delete” for a	Account permanently removed from the database; user no longer appears in donor search	Pass

Test Case ID	Test Case	Input	Expected Result	Status
	admin dashboard	selected user	results or admin listings	
S9	User logs out of the system	Authenticated user clicks "Logout"	Session terminated; user redirected to the login page; subsequent access to protected pages denied	Pass

Chapter 5: Conclusion and Future Recommendations

5.1 Conclusion

Sabaiko Blood Bank successfully demonstrates that a low-cost, open-source web platform can significantly reduce the time taken to locate compatible blood donors in Nepal. The implemented system is live, responsive, secure, and ready for pilot deployment.

5.2 Future Recommendations

- Integration with NRCS central database via API.
- Mobile applications (React Native) with push notifications.
- SMS gateway integration using local providers (Ncell/NTC API).
- AI-based prediction of blood demand during festivals/disasters.
- Blockchain-based immutable donation history for credibility.

Appendices (Screenshots)

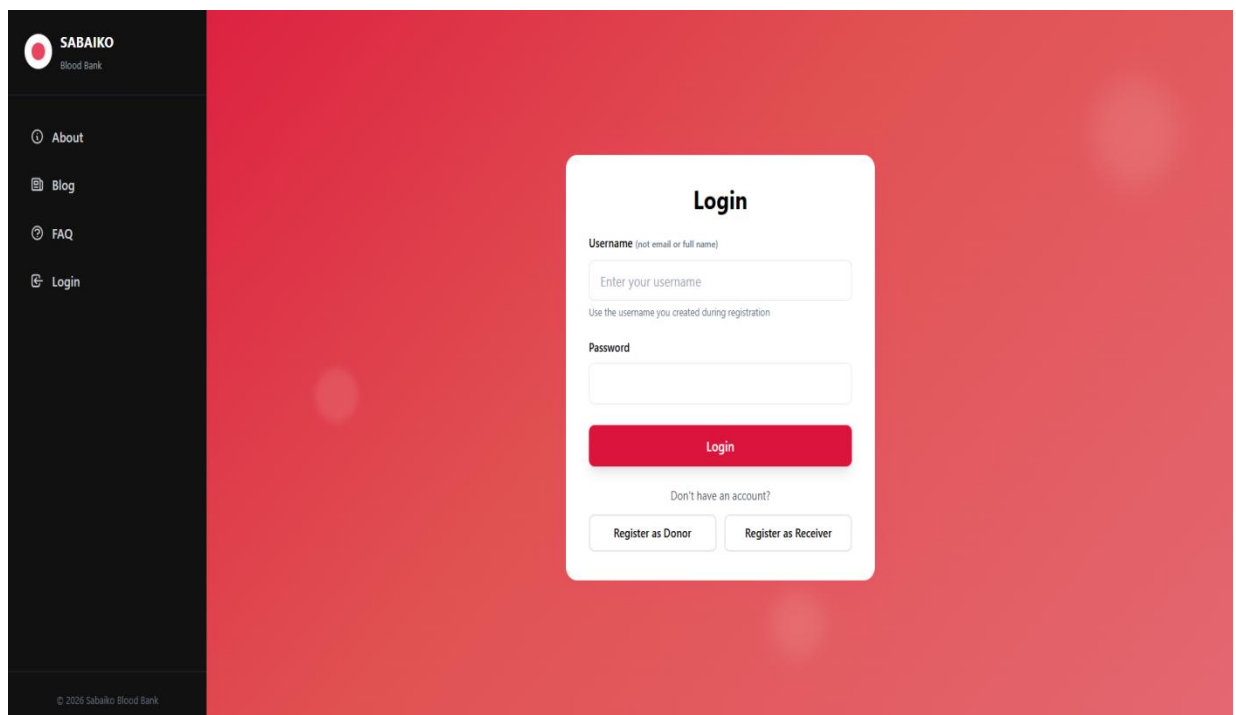


Figure 9: Login page

Register as Donor

Profile Image (Optional)

No file chosen

Username *

Email *

Password *

Full Name *

Phone *

Address Details *

Province + District + Municipality + Ward

Province

District

Municipality

Ward No.

Full Address: Not Provided

Address will be saved as a combination of the fields above.

Blood Group *

Select Blood Group



Register

Already have an account? [Login here](#)

Figure 10: Register as Donor

Register as Receiver

Profile Image (Optional)

No file chosen

Username *

Email *

Password *

Organization Name *

Phone *

PAN *

Address Details *

Province + District + Municipality + Ward

Province

District

Municipality

Ward No.

Full Address: Not Provided

Address will be saved as a combination of the fields above.

Register

Already have an account? [Login here](#)

Figure 11: Register as Receiver

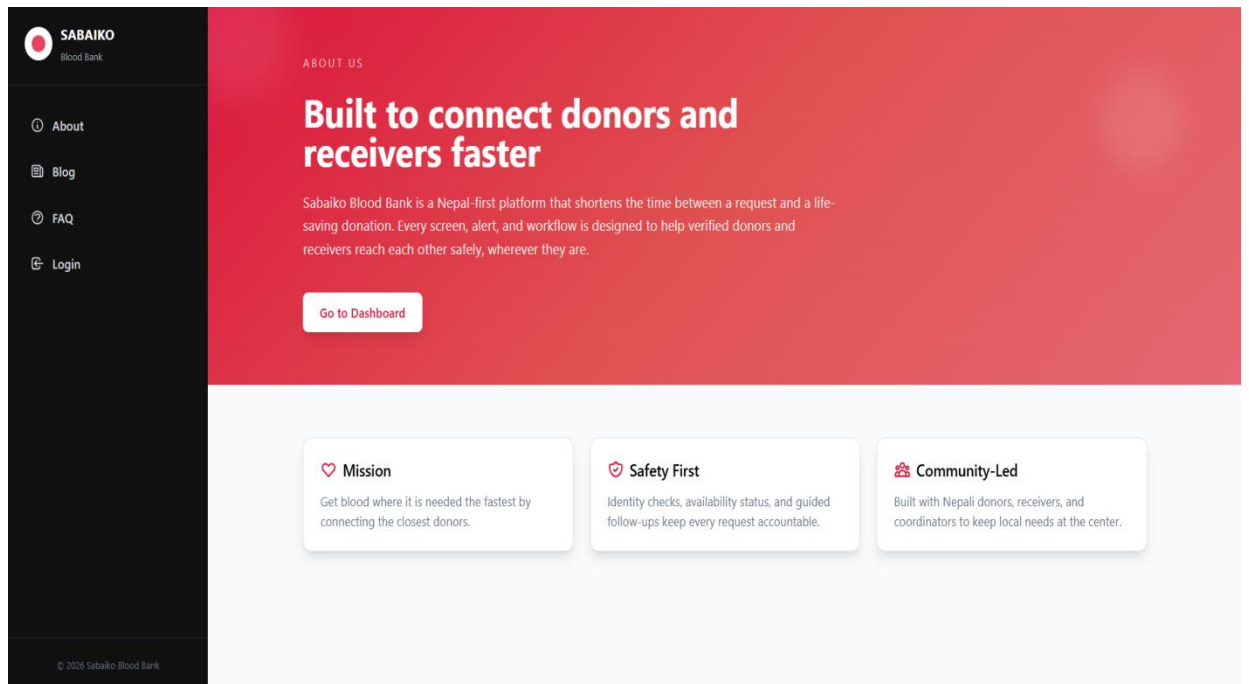


Figure 12: About page

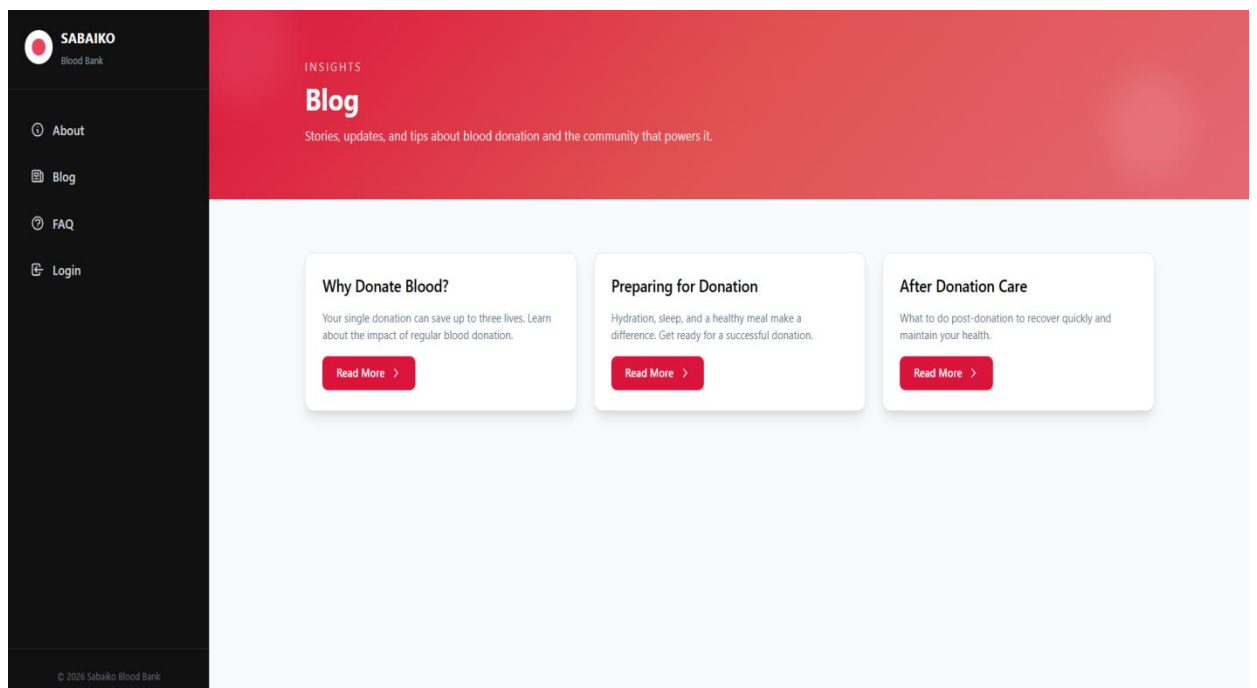


Figure 13: Blog page

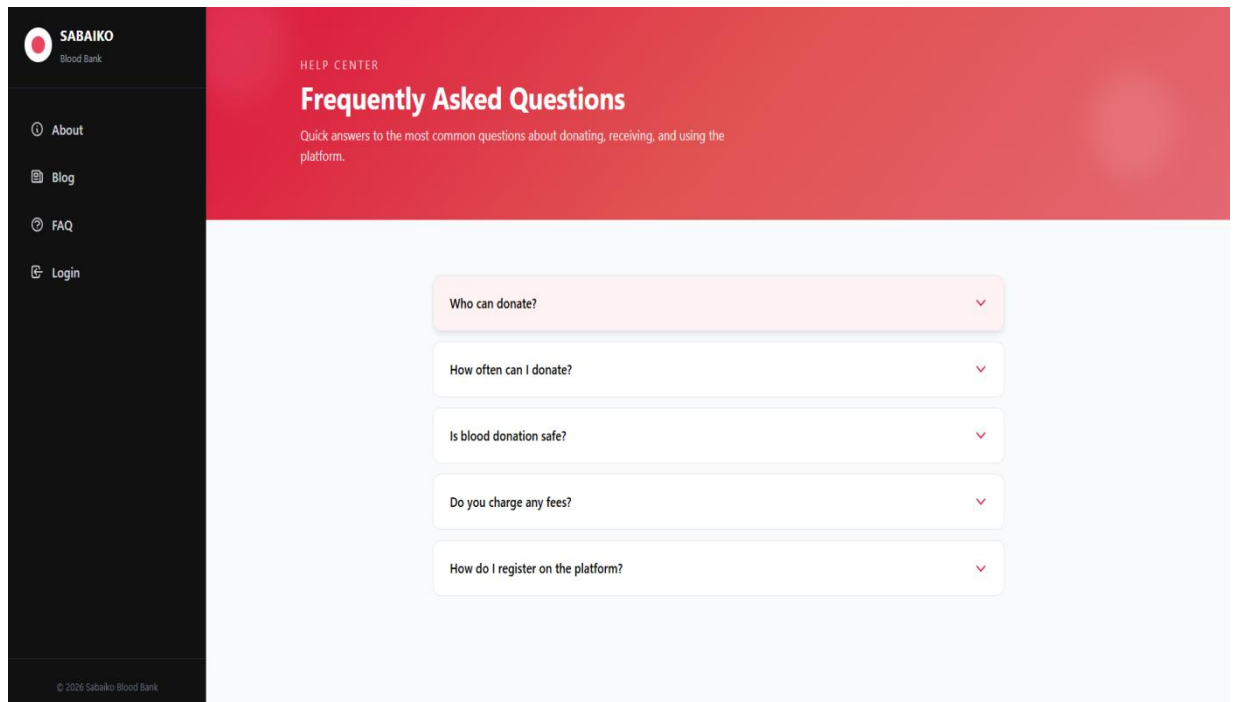


Figure 14: FAQ page

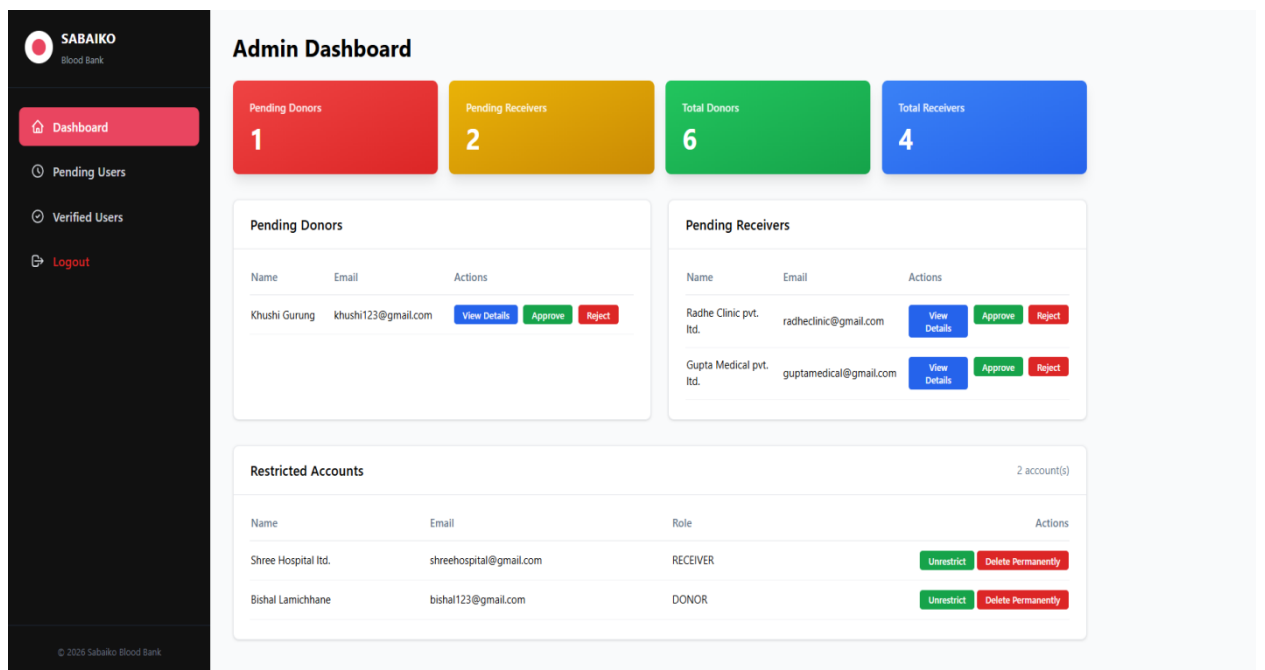


Figure 15: Admin Dashboard

SABAIIKO
Blood Bank

- Dashboard
- Pending Users
- Verified Users**
- Logout

© 2026 Sabaiko Blood Bank

Verified Users

Donors Receivers

NAME	EMAIL	BLOOD TYPE	ACTIONS
Sunil Bhattarai	sunil123@gmail.com	O-	View Details Restrict
Sita Shrestha	sita123@gmail.com	O+	View Details Restrict
Dinesh Poudel	dinesh123@gmail.com	O+	View Details Restrict
Sandhya Paudel	sandhya123@gmail.com	B+	View Details Restrict
Kishor Sharma	kishor123@gmail.com	A-	View Details Restrict
Niraj Baniya	niraj123@gmail.com	O-	View Details Restrict

Figure 16: Verified Donors

SABAIIKO
Blood Bank

- Dashboard
- Pending Users
- Verified Users**
- Logout

© 2026 Sabaiko Blood Bank

Verified Users

Donors Receivers

NAME	EMAIL	BLOOD TYPE NEEDED	STATUS	ACTIONS
Rama Medical pvt. Ltd.	ramamedical@gmail.com	Lumbini	✓ Verified	View Details Restrict
Red Cross Nepal	redcross@gmail.com	Bagmati	✓ Verified	View Details Restrict
Blood Bank Ltd.	bloodbank@gmail.com	Lumbini	✓ Verified	View Details Restrict
Pandey Clinic pvt. Ltd.	pandeyclinic@gmail.com	Gandaki	✓ Verified	View Details Restrict

Figure 17: Verified Receivers

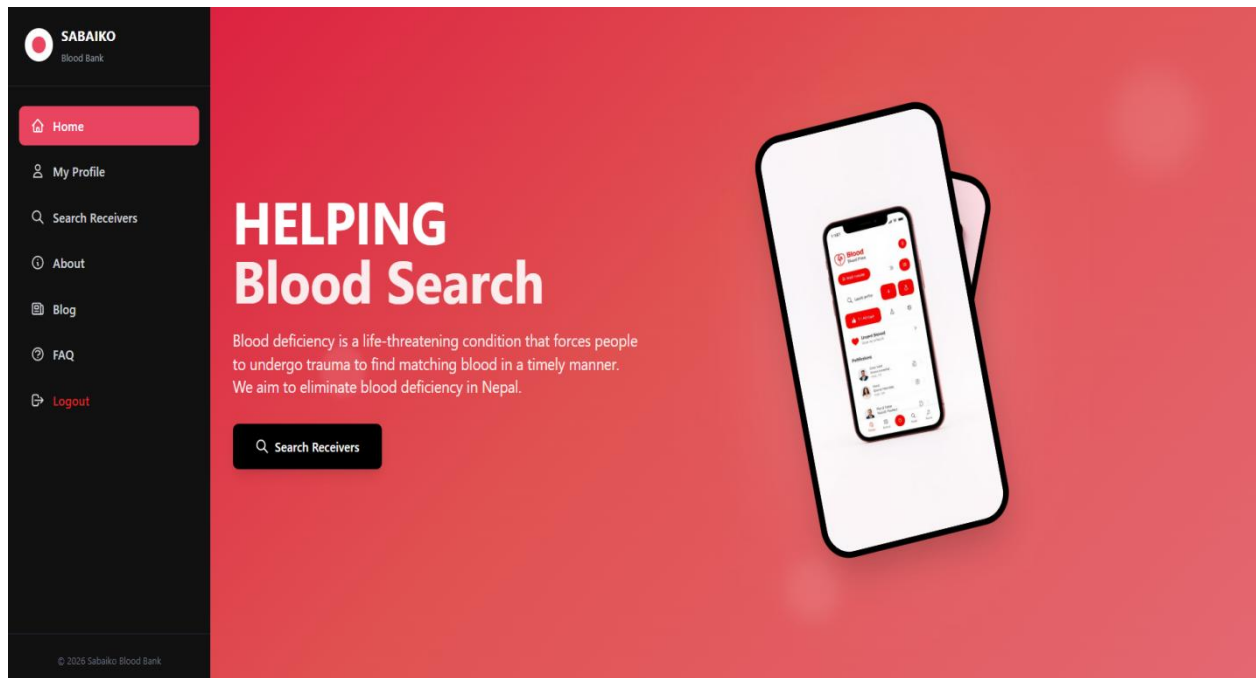


Figure 18: Donor Home page

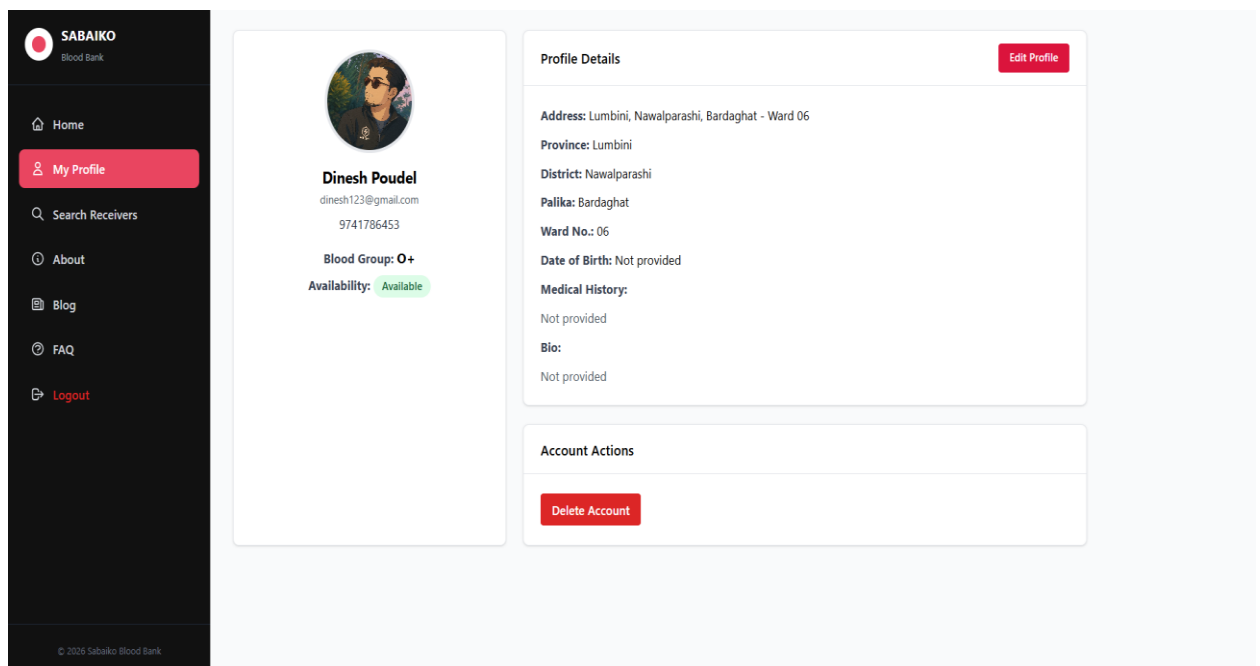


Figure 19: Donor Profile page

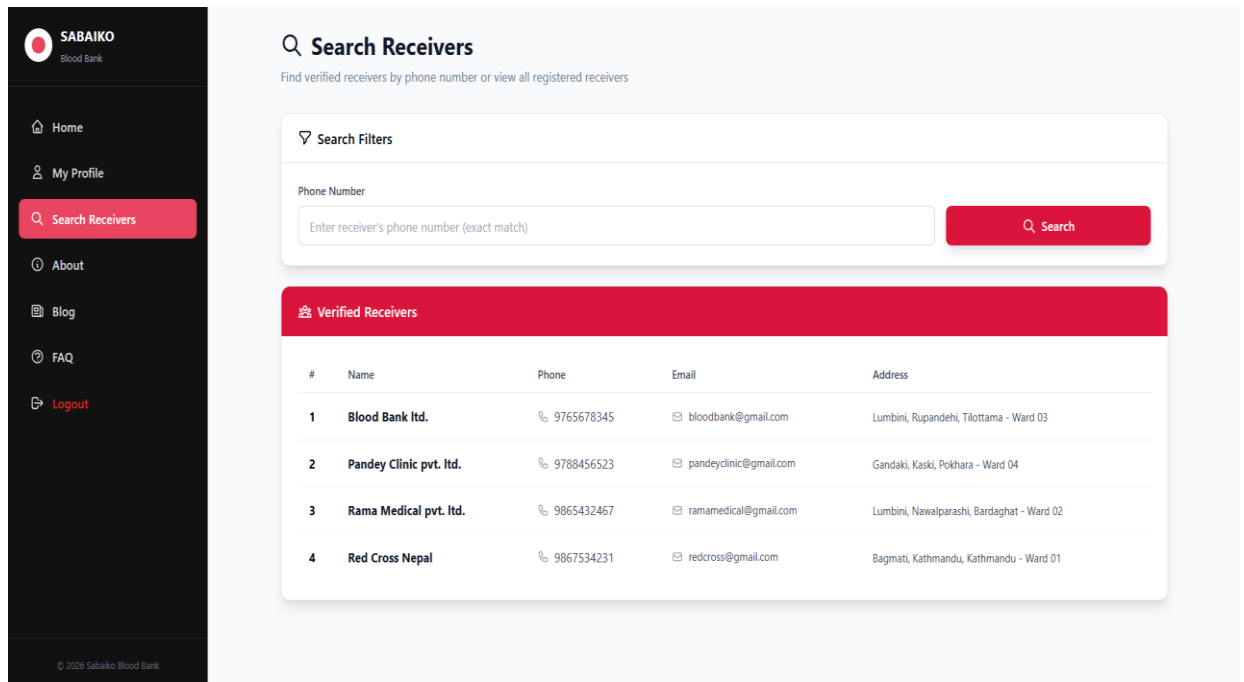


Figure 20: Search Receiver page

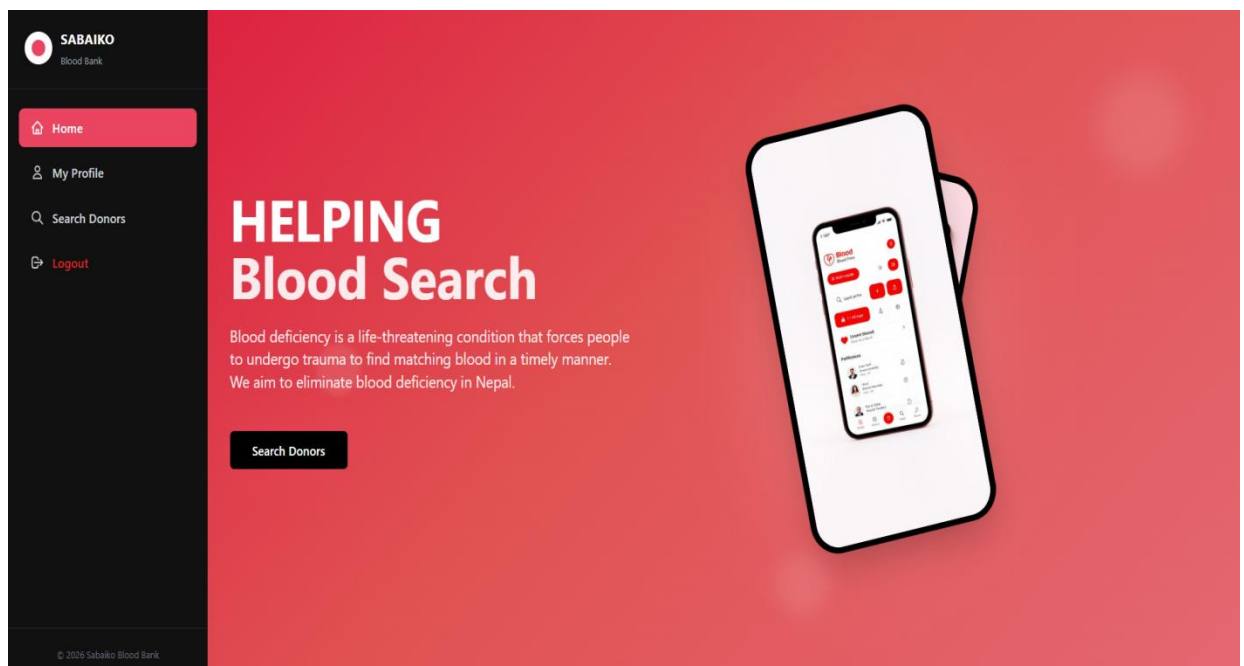


Figure 21: Receiver Home page

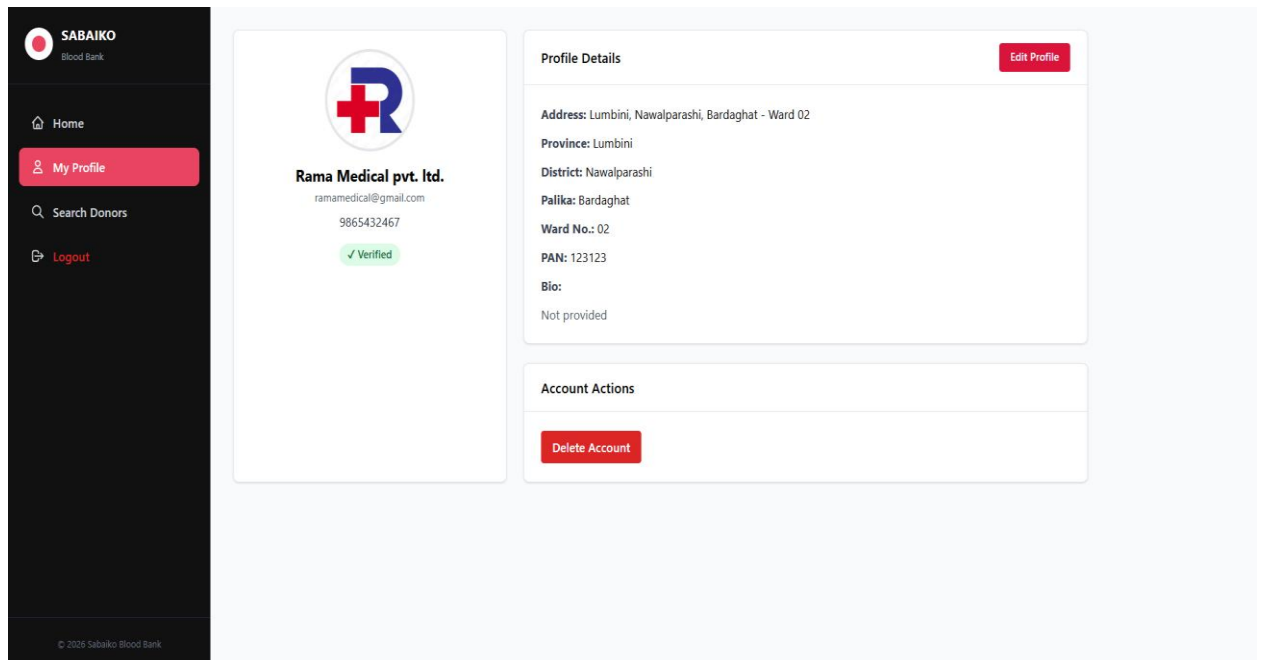


Figure 22: Receiver Profile page

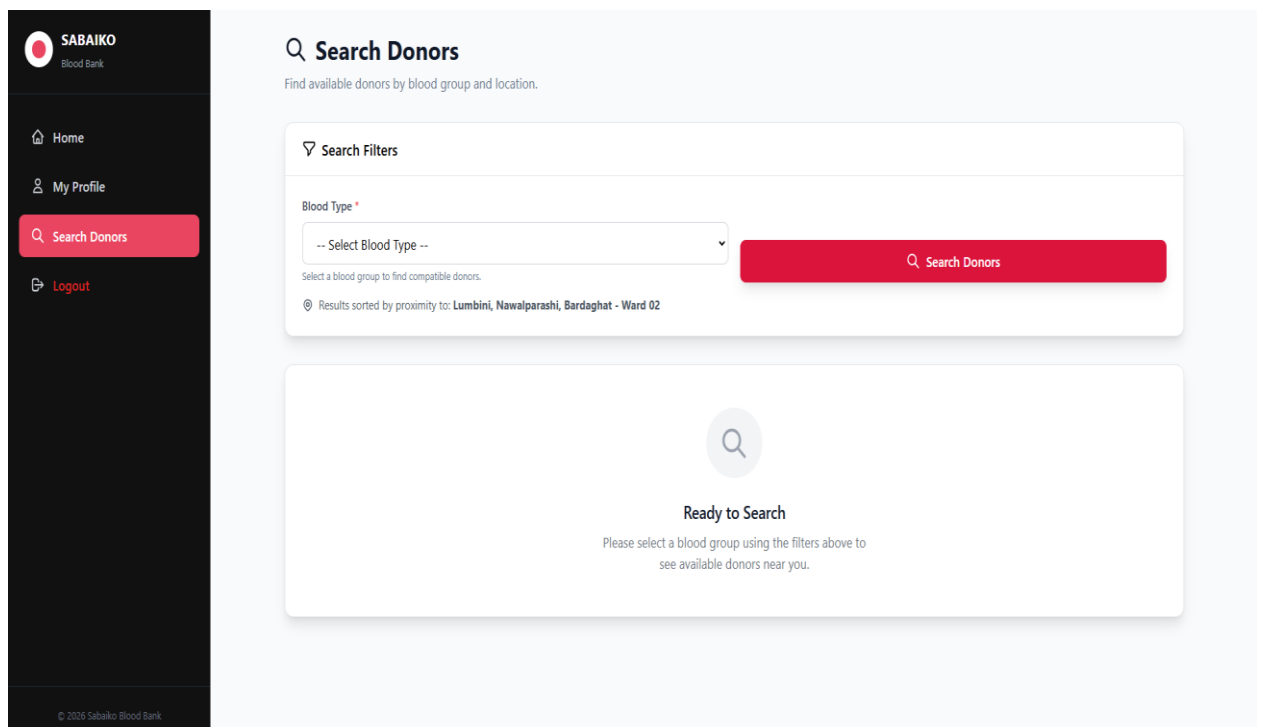


Figure 23: Search Donors

Supervisor Visit Log Sheet

SN	Date	Work Done	Supervisor Comment	Signature
1.	Nov 1	Project Kick-off		
2.	Nov 14	Requirements Gathering		
3.	Nov 29	System Design		
4.	Dec 30	Backend Development		
5.	Jan 14	Frontend Development		
6.	Jan 20	Testing & Debugging		
7.	Jan 22	Documentation		
8.	Feb 20	Presentation Prep		
9.	Feb 20	Review and Feedback		
10.	Feb 20	Conclusion		

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